Course Description: This course provides a survey of the skills and concepts needed for locating, managing, processing, and analyzing large amounts of data. Topics covered include data sourcing, choosing a big data infrastructure, extracting, transforming, loading, and mining large amounts of structured and unstructured data. The course will include hands-on experience working with sample technologies selected from a complex ecosystem of tools and platforms.

The Spring 2026 semester runs from January 20, 2026 through May 12, 2026. This class meets Tuesdays and Thursdays from 1:40 pm to 2:55 pm in EIT 220 as well as via a live webcast. All sessions are recorded.

Prerequisites: You are expected to have some basic proficiency in programming, databases, and statistics. Tutorials and review materials are available to help students fill in any gaps.

Course Objectives: By the end of this course, students will be able to:

- Compare and contrast the different approaches in data science for answering questions (*IFSC Graduate Program Objective 1 - Graduates will practice data-driven decision-making using current knowledge, skills, and technologies*).
- Describe the relative strengths and weaknesses of various scalable data analytics technologies and platforms (I*FSC Graduate Program Objective 1 - Graduates will practice data-driven decision-making using current knowledge, skills, and technologies*).
- Identify and articulate the underlying governance and ethical concepts associated with big data (*IFSC Graduate Program Objective 2 - Graduates will demonstrate the ability to critique and recommend improvements to Data Privacy/Security practices)*.
- Demonstrate written and oral communication skills needed to assume data project roles (*IFSC Graduate Program Objective 3 - Graduates will cultivate the communication and interpersonal skills needed to assume data project roles in a variety of settings (academia, government, and industry))*.
- Design and develop projects to investigate and answer questions using data (*IFSC Graduate Program Objective 4 - Graduates will design and develop projects to investigate and create solutions to information problems)*.

Course Structure: Each week will typically consist of activities such as lecture/discussion based on assigned readings, case studies involving the demonstration of a specific technology/tool, and hands-on lab exercises. Students will be asked to complete lab assignments based on the material presented in class and, in some cases, come prepared to discuss assigned readings. The reading assignments will come from recently published research and industry papers on big data topics.

Textbook: All materials will be on the course website and the cognitiveclass.ai training site. We will use a combination of relevant research papers, documentation from relevant systems, and readings from the web.

Cloud Computing: Options for cloud computing will be provided.

Other Resources: You will be expected to provide your own programming environment (e.g., a laptop.) If you need additional resources, let the instructor know.

Student Assessment: Important! The Data Science Project will receive 50% of the weight. The weekly exercises will receive the other 50% of the weight.

- 50% of your grade will be based on the points for completing a Data Science Project/Paper including a 3 Minute Talk presenting the Project/Paper Results. See the Project, Report and Presentation Guidelines in the Content Page for more details on the Project/Paper and Presentation requirements.
- 50% of your grade will be based on completing the weekly Assignments. Assignment points are based on the tutorials and weekly exercises completed by the student.
- Undergraduates in the course will receive a 10% bonus to make their grading level with the graduate students.

Grading Scale: 90% and above: A; 80-90% B; 70-80% C; 60-70% D, 60% and below: F.

Course Format / Instructor Presence

The course will be conducted as a lecture-based course. Participation counts and may include small group presentations and exercises. A high level of student participation is required. Make sure that all assigned preparation and readings are done in advance and that you are ready to engage in full examination and discussion of topics. The instructor will not hesitate to call on students for questions and comments. It is, therefore, critical that assignments be completed before class in which they are addressed and reviewed.

Attendance Expectations

- For In Person and Synchronous Online Students: Attendance is critical in mastering the course material. If you must miss class, please send an email to the instructor before class begins explaining the reason for your absence. Participation implies making comments, observations, contributions, and asking questions in the classroom.
- For Asynchronous Online Students: Keeping up with the recordings and weekly homework assignments is sufficient for establishing an active attendance record.

Communication Expectations for Regular and Substantive Interactions

In addition to the class sessions, the instructors for this course will have their weekly office hours posted in the Blackboard course shell. Course communications will be done via the Blackboard Announcements and the UALR email system. Students must use their UALR email when contacting their instructors. On weekdays, a student can expect a response to their email or phone message within twenty-four hours, and forty-eight hours on the weekends. Instructors also will initiate contact with the student, so it is essential that that the student monitor their UALR emails daily, review comments and feedback on assignments, and remain actively engaged no less than several times a week in their course(s). The student is responsible for knowing the syllabus, deadlines on assignments, school policies, and all communication from the instructors to him/her individually and to the class as a whole.

Standard Credit Hour Expectations

This course will adhere to the standard credit hour policy. For a 3-credit course, students can expect 2.5 hours per week of direct faculty instruction including in-class assessment and exams (midterms and final) and 5 hours per week of out-of-class time (e.g., studying, reading, completing assignments, working on projects, etc.).

Instructor's Late Work/Exam Make-Up Policy

Unless otherwise specified in class, work will be due on the date given on the class webpage or on the assignment itself. Late work will be accepted by the instructor at the instructor's discretion. Students who miss a scheduled presentation can petition the instructor for a make-up which may be granted at the instructor's discretion.

UA Little Rock Plagiarism/Academic Policies

Plagiarism on any assignment will at a minimum result in 0 points for the assignment. We reserve the right to pursue further disciplinary action if appropriate (e.g. any student caught cheating on an assignment/assessment will receive an "F" for the course, and we may pursue action with the Committee on Academic Integrity). Plagiarism includes copying someone else's work and claiming it as your own, or collaborating excessively with another person or persons and claiming the work as solely your own. It is strongly recommended that students maintain a record of the preparation of their major assignments. For more information on academic offenses, please refer to the following websites:

- **Academic Integrity and Grievance Policy:** https://ualr.edu/deanofstudents/section-vii-administration/academic-integrity-grievance-policy/
- **Academic Offenses:** https://ualr.edu/deanofstudents/section-vii-administration/academic-offenses/
- **Other Complaints:** https://ualr.edu/deanofstudents/student-complaints/

Inclement Weather Policy

During inclement weather, UA Little Rock will make a decision whether or not to close based on all available information. The chancellor will decide whether or not conditions warrant canceling classes and activities and closing the campus or whether classes and activities will be canceled but with specified campus offices open. Online or web-enhanced classes will continue as scheduled at the discretion of the faculty member. The UA Little Rock website, UA Little Rock email, the university's main telephone number (501.569.3000), and the Rave campus alert notification system are the official means of communicating information concerning weather-related closings. When necessary, the university will announce a separate decision about canceling night classes (those classes starting at 4:20 p.m. or later) by 2 p.m., if possible. For further information, Please review the Inclement Weather Policy available at https://ualr.edu/policy/home/admin/weather/.

UA Little Rock Disability Policy

If you are a student with a disability, or if you simply want to discuss resources that might help you learn more effectively, you can contact the Disability Resource Center at any time. You are welcome to drop in or call 501-916-3143 to make an appointment. Their staff will work with you to discuss accommodations and identify resources on campus or in the community that might be helpful for you. Accommodations are established through an interactive process. For more information, visit the DRC website or review the steps to request accommodations. We want you to know you are valued, welcome, and wanted at UA Little Rock.

Instruction Contact Information

Instructor # 1: **Dr. Serhan Dagtas,** Professor of Information Science

- Email: sxdagtas@ualr.edu
- Office: EIT 554
- Phone: 501-916-5227
- Office Hours: TBD

Instructor # 2: **Dr. Elizabeth Pierce**, Associate Professor and Chair of Information Science

- Email: expierce@ualr.edu
- Office: EIT 550
- Phone: 501-916-5223
- Office Hours: I am on campus Monday through Fridays, generally 9 am to 4:30 pm. I am available via email, phone, and by appointments.